

Experiment No. 8

Date: / /

Aim: To determine the enthalpy of neutralisation of strong acid 0.5M HCl and strong base 0.5M NaOH.

Theory: Enthalpy of neutralisation is an enthalpy change, when one mole of H^+ ions of an acid is completely neutralised by one mole of OH^- ions of a base to form one mole of water. Heat of neutralisation of any strong acid and strong base is almost constant, $\Delta H = -57.3 \text{ kJ}$. It is due to complete dissociation of strong acid and strong base.

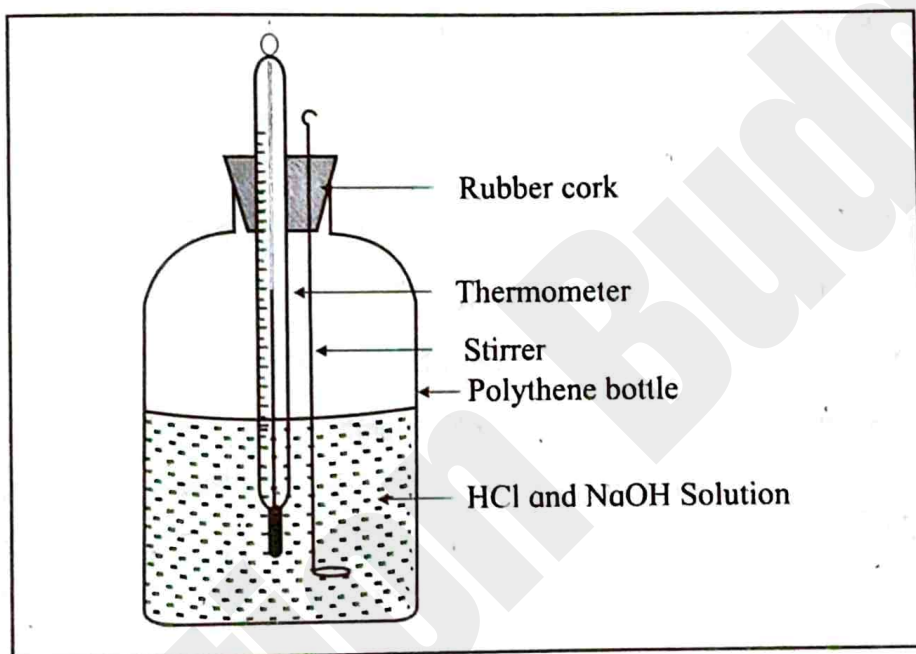
Reaction: $\text{NaOH(aq)} + \text{HCl(aq)} \longrightarrow \text{NaCl(aq)} + \text{H}_2\text{O(l)}; \Delta H = -57.3 \text{ kJ}$

In case of weak acid and strong base the enthalpy of neutralisation is less than that of strong acid and strong base as some heat is utilised for complete ionisation of weak acid.

Apparatus: Polythene bottle, cork with two holes, two beakers, thermometer ($1/10^\circ$), measuring cylinder, etc.

Chemicals: 0.5 M HCl and 0.5 M NaOH.

Diagram:



Procedure:

1. Take 50 mL 0.5 M HCl in a clean and dry polythene bottle.
2. Record its initial temperature ($t_1^\circ\text{C}$).
3. Wash the thermometer with water and gently wipe with filter paper.
4. Take 50 mL 0.5 M NaOH in a clean and dry beaker.
5. Record its initial temperature ($t_2^\circ\text{C}$).
6. Add NaOH solution to the polythene bottle, shake well.
7. Record the constant maximum temperature of the mixture ($t_4^\circ\text{C}$)

Note :

- i. Specific heat capacity of commonly used PET type of polythene bottle can be approximated to $1 \text{ J/g } ^\circ\text{C} = 1000 \text{ J/kg } ^\circ\text{C} \approx 0.25 \text{ cal/g } ^\circ\text{C}$
- ii. If the bottle is thin walled, its mass can be neglected if mass of water inside it is more than 100g.
- iii. If a heavy bottle of different material is used, we need to know and use its water equivalent.

Observation Table:

1	Initial temperature of 0.5 M HCl	t_1	25.....°C
2	Initial temperature of 0.5 M NaOH	t_2	25.....°C
3	Mass of solution = Total volume of solution	m	100 g
4	Average temperature of both solutions (HCl + NaOH)	$t_3 = \frac{25(t_1) + 25(t_2)}{2}$	25.....°C
5	Final temperature of mixture	t_4	28.....°C
6	Rise in temperature	$\Delta t = 28^\circ\text{C} (t_4) - 25^\circ\text{C} (t_3)$	3.....°C
7	Specific heat capacity of water	S	4.184 J/g °C

Calculation:

Enthalpy change during neutralisation of 50 mL 0.5 M HCl by 50 mL 0.5M NaOH.

$$Q = m S \Delta t$$

$$Q = 100 \times 4.184 \text{ J/g } ^\circ\text{C} \times 3 \Delta t$$

$$Q = 1255.2 \text{ J}$$

1000 mL 0.5 M HCl contain 0.5 mol of HCl

50 mL 0.5 M HCl will contain 0.025 mol of HCl

..... (Q) heat evolved when 0.025 mol of HCl neutralised

Enthalpy of neutralisation is therefore

$$\Delta H = \frac{1255.2 \text{ (Q)}}{0.025}$$

$$\Delta H = -50208 \text{ J/mol}$$

$$\Delta H = -50.208 \text{ kJ/mol}$$

Space for log calculation

Number	log
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Result:-

The Enthalpy of neutralisation of strong acid HCl and strong base NaOH = kJ/mol
(Negative sign is because reaction is exothermic in nature)

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

1. Chemical reaction that absorbs heat from the surrounding is said to be and has ΔH at constant pressure
 - a. exothermic, positive
 - b. exothermic, negative
 - c. endothermic, negative
 - ☒ d. endothermic, positive
2. The following apparatus is used to determine the enthalpy of neutralisation of a given salt
 - a. viscometer
 - ☒ b. calorimeter
 - c. potentiometer
 - d. hygrometer
3. The pH of soil can be controlled through
 - a. neutralisation
 - b. acidic oxides
 - c. neutral oxides
 - ☒ d. basic oxides
4. When an acid is (H^+) added to alkali (OH^-) product is
 - a. hydroxide
 - ☒ b. water
 - c. salt
 - d. H_2 gas

Short answer questions

1. What is the enthalpy of neutralisation of a strong acid and a strong base?

Ans. The value of enthalpy of neutralisation of strong acid and strong base is -50.3 kJ

2. Write the following statement as True/false

1. An exothermic reaction is characterized by the positive value of ΔH false
2. Heat is evolved when an exothermic reaction occurs true
3. An endothermic reaction is characterized by the positive value of ΔH true
4. The reaction vessel becomes hot when an endothermic reaction occurs false

3. Define enthalpy of neutralisation.

Ans. Enthalpy of neutralisation is defined as enthalpy change when one mole of H^+ ions of acid is completely neutralised by one mole of OH^- ions of base.

4. Why the enthalpy of neutralisation of a strong acid with strong base is always the same?

Ans. The enthalpy of neutralisation of strong acid with strong base is always same (i.e. -50.3) kJ due to complete dissociation of strong acid and strong base.

Remark and sign of teacher: